AI Engineer

Sample Job Description

Company: NovaTech AI Solutions

Location: Bengaluru, India (Hybrid)

Employment Type: Full-time

Experience: 0–3 years

About the Role

We are looking for an AI Engineer to design, build, evaluate, and deploy AI-powered applications. The role involves working with large language models, retrieval-augmented generation (RAG), machine learning pipelines, APIs, and production software systems. You will work closely with software engineers, data scientists, and product teams to turn AI capabilities into reliable user-facing products.

Key Responsibilities

- Develop and deploy AI-powered applications using Python and modern AI/ML frameworks.
- Build LLM-based applications including conversational assistants, RAG systems, and tool-using agents.
- Design document ingestion, chunking, embedding, retrieval, reranking, and context-generation pipelines.
- Integrate commercial and open-source LLMs through APIs and model-serving frameworks.
- Develop REST APIs and backend services using frameworks such as FastAPI or Flask.
- Evaluate LLM and RAG systems using relevant quality, latency, reliability, and cost metrics.
- Implement prompt engineering, structured outputs, guardrails, and basic AI safety practices.
- Work with vector databases and search technologies such as FAISS, Qdrant, Pinecone, or similar tools.
- Containerize applications using Docker and support deployment to cloud environments.
- Write maintainable, testable code and collaborate through Git-based development workflows.

Required Skills

- Strong Python programming fundamentals and familiarity with object-oriented programming.
- Understanding of machine learning fundamentals, including supervised learning, model evaluation, and data preprocessing.
- Hands-on understanding of LLM concepts, prompting, embeddings, vector search, and RAG.
- Experience building applications with LLM APIs such as OpenAI, Anthropic, Groq, or equivalent providers.
- Knowledge of REST APIs, JSON, HTTP, and backend development.
- Familiarity with Git and GitHub.
- Basic SQL and database concepts.
- Understanding of Docker and containerized application deployment.

Preferred Skills

- Experience with LangChain, LangGraph, LlamaIndex, or equivalent AI application frameworks.
- Experience with Qdrant, Pinecone, Weaviate, FAISS, or another vector database/search system.
- Experience with FastAPI, Streamlit, React, or similar application frameworks.
- Familiarity with AWS, Google Cloud, Azure, Railway, or other cloud platforms.
- Knowledge of CI/CD, logging, monitoring, observability, and application security.
- Exposure to fine-tuning, LoRA/PEFT, model quantization, or local LLM inference.

Education

Bachelor's degree in Computer Science, Artificial Intelligence, Data Science, Engineering, Mathematics, or a related technical field. Equivalent practical experience and strong project work may also be considered.

Example Projects

- Production-ready RAG chatbot with conversation history and source-aware retrieval.
- AI resume/job-description matching system using embeddings and LLM-based analysis.
- Tool-using AI agent connected to external APIs and databases.
- Document question-answering system supporting PDF, DOCX, and TXT files.

What We Look For

We value strong programming fundamentals, practical problem-solving, curiosity, ability to learn new AI technologies, and the ability to turn prototypes into reliable software. Candidates should be able to explain their technical decisions and demonstrate working projects through a portfolio or GitHub.